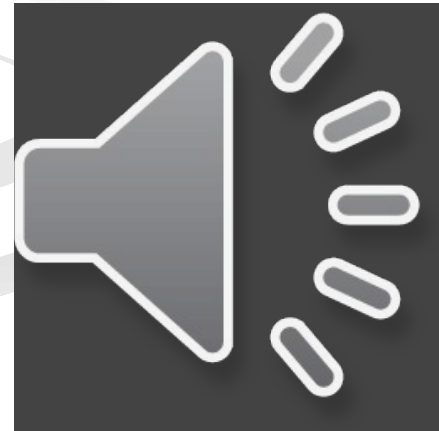


Meta-analysis

**Master of Statistics
KU Leuven**

Data extraction



Steps in a meta-analysis

1. Formulating research questions
2. Systematic search for studies
3. Selection of relevant studies
4. Extracting relevant data (outcomes, study characteristics, study quality)
5. Converting data to a common effect size measure
6. Combining and comparing the effect sizes
7. Looking for moderator variables
8. Concluding and reporting



Extracting data

- The Effect Size
 - effect sizes, summary data, test statistics, p -values
 - confidence about the size of the effect: standard error, sample size, test statistics, p -values
- Information about the study (moderator variables!):
 - characteristics of participants
 - operationalization of the dependent variable & treatment
 - contextual elements (location, time, ...)
 - statistical tests
 - the design
 - study quality
 - ...



Data management

- Use a coding scheme
- One sheet for each study (or part of the study)
- Clearly indicate reference and pages
- Better to code somewhat too much than too little, but not too much!



[**VARIABLE NAMES IN BRACKETS**]

Bibliographic reference: _____

- — — — 1. Study ID number [STUDYID]
— 2. Type of publication [PUBTYPE]
1 book 4 technical report
2 journal article 5 conference paper
or book chapter 6 other (specify): _____
3 thesis or doctoral
— — 3. Publication year (last two digits; 99 if unknown)?
[PUBYEAR]

Sample Descriptors

- — . —
-
4. Mean age [MEANAGE]
5. Predominant race [RACE]
- | | |
|-----------------------|-------------------------------------|
| 1 >60% White | 5 mixed, none more than 60% |
| 2 >60% Black | 6 mixed, cannot estimate proportion |
| 3 >60% Hispanic | |
| 4 >60% other minority | 9 cannot tell |
-
6. Predominant sex [SEX]
- | |
|---------------|
| 1 <5% male |
| 2 5%–50% male |
| 3 50% male |



Sample Size

- — — 11a. Treatment group sample size [TXN]
- — — 11b. Control group sample size [CGN]

Means and Standard Deviations

- — — — 12a. Treatment group mean [TXMEAN]
- — — — 12b. Control group mean [CGMEAN]
- — — — 13a. Treatment group standard deviation [TXSD]
- — — — 13b. Control group standard deviation [CGSD]

Proportions or Frequencies

- — — — 14a. n of treatment group with a successful outcome [TXSUCCES]
- — — — 14b. n of control group with successful outcome [CGSUCCES]
- — — — 15a. Proportion of treatment group with a successful outcome [TXPROP]
- — — — 15b. Proportion of control group with a successful outcome [CGPROP]



Data management

	A	B	C	D
1	study_id	Year	PubType	Pub
2	10001	2012	1	1
3	10003	2012	2	1
4	10004	2006	1	1
5	10006	2006	1	1
6	10007	2000	1	1
7	10011	1999	1	1
8	10012	1998	1	1
9	10014	1990	1	1

	A	B	C	D	E	F	G	H	I	J	K	L
1	study_id	sample_id	Country	NoPart	NoMale	NoFemale	%Male	%Female	AgeMean	AgeSD	AgeRange	AgeCat
2	10001	1	BE	216	166	50	0.768519	0.231481				2
3	10003	1	IT+UK+ES	5862	2986	2847	0.509382	0.48567			12-16	2
4	10004	1	IT	330	162	168	0.490909	0.509091	15.074	2.47	11-19	2
5	10006	1	AR	1019	483	536	0.473994	0.526006			8-12	1
6	10007	1	IE	1090	396	688	0.363303	0.631193	15.63229		13-18	2
7	10011	1	BE	370	175	195	0.472973	0.527027	13.73568	2.547685	11-17	2
8	10012	1	BE	200	106	94	0.53	0.47	14.89			2
9	10012	2	BE	358	170	188	0.47486	0.52514				2
10	10014	1	BE	338	168	169	0.497041	0.5			10-17	2
11	10017	1	CA	644	350	294	0.543478	0.456522	15.78	1.38	13-18	2
12	10018	1	IT	522	215	307	0.411877	0.588123	15.6	0.98	14-17	2
13	10018	2	IT	209	87	122	0.416268	0.583732	16.5	0.96	15-19	2
14	10020	1	UK	296	146	150	0.493243	0.506757	9			1
15	10021	1	IL	231	103	128	0.445887	0.554113	17.04	0.3		2



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	id	id2	outcome	Scale	Scale2	NameQ	NoItems	NoCat	RangeCat	Mean	SD	Scale	Alpha
2	10001	1	1	LACA_LPEERS	12	Loneliness and Alonene	6	4	1-4	10.66	4.26	2	0.87
3	10003	1	2	LACA_LPART	11	Louvain Loneliness Scal	3	4	1-4	1.41	0.58	1	0.83
4	10003	1	3	LACA_LPEERS	12	Louvain Loneliness Scal	3	4	1-4	1.54	0.64	1	0.78
5	10004	1	4	LACA_LPART	11	Louvain Loneliness Scal	12	4	1-4	1.7550303	0.5546403	1	
6	10004	1	5	LACA_LPEERS	12	Louvain Loneliness Scal	12	4	1-4	1.7414545	0.5414238	1	
7	10006	1	6	LACA_LPART	11	Louvain Loneliness Scal	4	4	1-4				0.7
8	10006	1	7	LACA_LPEERS	12	Louvain Loneliness Scal	4	4	1-4				0.72
9	10007	1	8	LACA_LPART	11	Louvain Loneliness Scal	12	4	1-4	24.761835	11.63136	2	
10	10007	1	9	LACA_LPEERS	12	Louvain Loneliness Scal	12	4	1-4	24.268991	14.231359	2	
11	10011	1	10	LACA_LPART	11	Louvain Loneliness Scal	12	4	1-4	19.917703	1.6382734	2	0.88
12	10011	1	11	LACA_LPEERS	12	Louvain Loneliness Scal	12	4	1-4	21.332486	2.0800459	2	0.86
13	10012	1	12	LACA_LPART	11	Louvain Loneliness Scal	12	4	1-4	18.5108		2	
14	10012	1	13	LACA_LPEERS	12	Louvain Loneliness Scal	12	4	1-4	21.06595		2	
15	10012	2	14	LACA_LPART	11	Louvain Loneliness Scal	12	4	1-4	20.296648		2	

Maes, M., Van Den Noortgate, W., Goossens, L. (2015). A Reliability Generalization Study for a Multidimensional Loneliness Scale: The Loneliness and Aloneness Scale for Children and Adolescents. *European Journal of Psychological Assessment*, 31 (4), 294-301. [027/1015-5759/a000237](https://doi.org/10.1027/1015-5759/a000237) Open Access



Inter- & intracoder reliability

- Unambiguous coding scheme (internal reliability or consistency).
- Report important choices (external reliability or replicability)
- Coders should be trained
- Preferably (partly) coded by two independent coders
- Report reliability for each characteristic separately:
 - E.g., effect size, N, design, quality, ...

